Newton on KKT	KKT Analytical
min $f(x) = x_1^2 + 5x_2^2$, s.t. $g(x) = 4 - x_1 - x_2 \le 0$	min $f(x) = x_1^2 + 5x_2^2$, s.t. $g(x) = x_1 + x_2 - 4 \le 0$
	Slack: $g(x) - s^2 = 0$, so $l(x, \lambda, s) = f(x) - \lambda(g(x) - s^2)$
	$\dfrac{\partial l}{\partial s} = 2\lambda s = 0 \ \Rightarrow\ \lambda = 0$ (inactive) or $s = 0$ (active)
	Test: $X_{\text{unc}} = (0,0)$; $g(X_{\text{unc}}) = -4 < 0 \quad \therefore$ infeasible $\Rightarrow \lambda \ne 0$ $\xRightarrow{2\lambda s=0}$ $s = 0$, constraint **active**
$l(x,\lambda) = x_1^2 + 5x_2^2 - \lambda(x_1 + x_2 - 4)$ (same l, implicit in $\nabla_x l$)	$l(x,\lambda) = x_1^2 + 5x_2^2 - \lambda(x_1 + x_2 - 4)$ (set $s = 0$)
Assert $r(x,\lambda) = 0$ and J; note $J = \begin{bmatrix} H_l & -\nabla_x g \\ \nabla_x g^\top & 0 \end{bmatrix}$: $r = \begin{bmatrix} \nabla_x l \\ g(x) \end{bmatrix} = \begin{bmatrix} 2x_1 - \lambda \\ 10x_2 - \lambda \\ x_1 + x_2 - 4 \end{bmatrix} \quad J = \begin{bmatrix} 2 & 0 & -1 \\ 0 & 10 & -1 \\ 1 & 1 & 0 \end{bmatrix}$	Same r and J arrived at from $\nabla_x l = 0$ and $g = 0$: $r = \begin{bmatrix} \nabla_x l \\ g(x) \end{bmatrix} = \begin{bmatrix} 2x_1 - \lambda \\ 10x_2 - \lambda \\ x_1 + x_2 - 4 \end{bmatrix} \quad J = \begin{bmatrix} 2 & 0 & -1 \\ 0 & 10 & -1 \\ 1 & 1 & 0 \end{bmatrix}$
Init: $z_0 = (0,0,0)^\top$; evaluate r at z_0: $r(z_0) = \begin{bmatrix} 2x_1 - \lambda \\ 10x_2 - \lambda \\ x_1 + x_2 - 4 \end{bmatrix}_{z_0=(0,0,0)} = \begin{bmatrix} 2(0) - 0 \\ 10(0) - 0 \\ 0 + 0 - 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -4 \end{bmatrix}$	(no iterate — constraint known active from test above)
Newton step: $J\,\Delta z = -r(z_0)$, i.e. $-r(z_0) = (0,0,4)^\top$: $J\,\Delta z = \begin{bmatrix} 2 & 0 & -1 \\ 0 & 10 & -1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \\ \Delta\lambda \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 4 \end{bmatrix}$ Row 1: $2\Delta x_1 - \Delta\lambda = 0 \ \Rightarrow\ \Delta x_1 = \frac{\Delta\lambda}{2}$ Row 2: $10\Delta x_2 - \Delta\lambda = 0 \ \Rightarrow\ \Delta x_2 = \frac{\Delta\lambda}{10}$	$\nabla_x l = H_f\, x - \lambda\,\nabla_x g = 0 \ \Rightarrow\ x^*(\lambda) = H_f^{-1}\lambda\,\nabla_x g$: $\begin{bmatrix} 2x_1 - \lambda \\ 10x_2 - \lambda \end{bmatrix} = 0 \ \Rightarrow\ x^*(\lambda) = \begin{bmatrix} \lambda/2 \\ \lambda/10 \end{bmatrix}$
Row 3: $\frac{\Delta\lambda}{2} + \frac{\Delta\lambda}{10} = 4 \ \Rightarrow\ 6\Delta\lambda = 40 \ \Rightarrow\ \Delta\lambda = \frac{20}{3}$	Sub $x^*(\lambda)$ into $g = 0$: $\frac{\lambda}{2} + \frac{\lambda}{10} = 4 \ \Rightarrow\ 10\lambda + 2\lambda = 80 \ \Rightarrow\ \lambda = \frac{20}{3}$
$\Delta x_1 = \dfrac{10}{3}$, $\Delta x_2 = \dfrac{2}{3}$; $z_1 = z_0 + \Delta z = \left(\frac{10}{3}, \frac{2}{3}, \frac{20}{3}\right)$	$x^* = \left(\dfrac{10}{3}, \dfrac{2}{3}\right)^\top$
$f(x^*) = \left(\frac{10}{3}\right)^2 + 5\left(\frac{2}{3}\right)^2 = \frac{100}{9} + \frac{20}{9} = \frac{40}{3}$	$f(x^*) = \left(\frac{10}{3}\right)^2 + 5\left(\frac{2}{3}\right)^2 = \frac{100}{9} + \frac{20}{9} = \frac{40}{3}$

SQP connection

KKT — necessary conditions for a constrained optimum; gives the system $r(x,\lambda) = 0$

Lagrangian — function whose stationarity encodes the KKT conditions; introduces λ

Slack variables — convert inequalities to equalities; lead to complementary slackness $\lambda s = 0$

SQP — solves $r = 0$ using Newton's method; each step inverts the Jacobian of r:

$$r(x,\lambda) = \begin{bmatrix} \nabla_x \mathcal{L} \\ g(x) \end{bmatrix} = \begin{bmatrix} 2x_1 - \lambda \\ 10x_2 - \lambda \\ x_1 + x_2 - 4 \end{bmatrix} = 0 \qquad J = \frac{\partial r}{\partial(x_1, x_2, \lambda)} = \begin{bmatrix} 2 & 0 & -1 \\ 0 & 10 & -1 \\ 1 & 1 & 0 \end{bmatrix}$$

BFGS — quasi-Newton approximation of the Hessian of $\mathcal{L}$ when exact computation is too costly

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Newton's method applied to KKT (same example: min $x_1^2 + 5x_2^2$ s.t. $x_1 + x_2 = 4$, starting at $z_0 = (0,0,0)$)

Step 1 — residual at z_0:

$$r(z_0) = \begin{bmatrix} 2(0) - 0 \\ 10(0) - 0 \\ 0 + 0 - 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -4 \end{bmatrix}$$

Step 2 — Jacobian (constant here since r is linear):

$$J = \begin{bmatrix} 2 & 0 & -1 \\ 0 & 10 & -1 \\ 1 & 1 & 0 \end{bmatrix}$$

Step 3 — solve $J\,\Delta z = -r(z_0)$:

$$\begin{bmatrix} 2 & 0 & -1 \\ 0 & 10 & -1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \\ \Delta\lambda \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 4 \end{bmatrix}$$

Row 1: $2\Delta x_1 - \Delta\lambda = 0 \ \Rightarrow\ \Delta x_1 = \frac{\Delta\lambda}{2}$
Row 2: $10\Delta x_2 - \Delta\lambda = 0 \ \Rightarrow\ \Delta x_2 = \frac{\Delta\lambda}{10}$
Row 3: $\Delta x_1 + \Delta x_2 = 4 \ \Rightarrow\ \frac{\Delta\lambda}{2} + \frac{\Delta\lambda}{10} = 4 \ \Rightarrow\ 6\Delta\lambda = 40 \ \Rightarrow\ \Delta\lambda = \frac{20}{3}$

$$\Delta x_1 = \frac{10}{3} \qquad \Delta x_2 = \frac{2}{3}$$

Step 4 — update: $z_1 = z_0 + \Delta z$

$$z_1 = \left(\frac{10}{3}, \frac{2}{3}, \frac{20}{3}\right) = x^* \quad \checkmark \quad \text{(one step: KKT system is linear)}$$